

ECOM30001/90001 Basic Econometrics

Tutorial 7: Binary Choice Models (LPM vs. Probit)

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Tutorial 7 Roadmap & Key Questions

Today's Core Econometric Goals

- **Binary Dependent Variables:** Why ordinary OLS fails and what the Linear Probability Model (LPM) assumes.
- **LPM Margins & Turning Points:** Interpreting continuous slopes, dummy differences, and quadratic tenure dynamics.
- **The Probit Model:** Non-linear link functions $\Phi(I_i)$, index functions, and bounded probabilities in $(0, 1)$.
- **Marginal Effects in Probit:** Calculating calculus-based Average Marginal Effects (AME) vs. discrete differences for dummies.
- **R Practice:** `lm()`, `glm(..., family=binomial(link="probit"))`, and `margins`.

Empirical Setting: Telecom Customer Churn

Econometric Latent Variable Model (1)

$$\begin{aligned}\text{churn}_i^* = & \beta_0 + \beta_1 \text{partner}_i + \beta_2 \text{dependents}_i + \beta_3 \text{tenure}_i + \beta_4 \text{tenure}_i^2 \\ & + \beta_5 \text{monthlycharges}_i + \beta_6 \text{contract2}_i + \beta_7 \text{contract3}_i + \beta_8 \text{paperless}_i + \varepsilon_i\end{aligned}$$

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Variables in telecom.csv ($n = 7,043$):

- churn^* = latent variable determining whether a customer switched providers in the last month ($\text{churn}_i = 1$ if $\text{churn}_i^* \geq 0$, 0 otherwise)
- $\text{partner} = 1$ if customer has a partner, 0 otherwise
- $\text{dependents} = 1$ if customer has dependents, 0 otherwise
- $\text{tenure} =$ number of months customer has stayed, 0 otherwise
- $\text{monthlycharges} =$ monthly account charges, in dollars
- $\text{contract1} = 1$ if customer does not have a contract ('month-to-month'), 0 otherwise (**omitted category**)
- $\text{contract2} = 1$ if customer has a 12-month contract, 0 otherwise
- $\text{contract3} = 1$ if customer has a 24-month contract, 0 otherwise
- $\text{paperless} = 1$ if customer receives bills electronically, 0 otherwise

Warm-Up Intuition Check

Quick Discussion (1 minute with your neighbor)

Before looking at regression estimates, what signs do you expect for:

- ① $\text{monthlycharges}_i \rightarrow$ Positive or Negative?
- ② contract2_i and $\text{contract3}_i \rightarrow$ Positive or Negative?
- ③ $\text{tenure}_i \rightarrow$ Does churn probability monotonically drop forever?

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- 1 monthlycharges_i $\rightarrow$ Positive or Negative?
- 2 contract2_i and contract3_i $\rightarrow$ Positive or Negative?
- 3 tenure_i $\rightarrow$ Does churn probability monotonically drop forever?

Economic Rationale:

- monthlycharges $\uparrow \implies$ **Positive sign** (higher price increases switching incentive).
- Long-term contracts $\implies$ **Negative sign** (switching costs / penalties).
- tenure may be **U-shaped**: loyalty drops early risk, but long-tenured customers may eventually shop around or face outdated plans.

Table 1: LPM and Probit Estimates

Variable	OLS (LPM, Robust SE)	Probit
Intercept	0.3265*** (0.0152)	−0.8288*** (0.0580)
partner	0.0097 (0.0111)	0.0431 (0.0441)
dependents	−0.0413*** (0.0108)	−0.1716*** (0.0491)
tenure	−0.0119*** (0.0008)	−0.0442*** (0.0032)
tenure ²	0.0001*** (0.0000)	0.0004*** (0.0000)
monthly charges	0.0032*** (0.0002)	0.0152*** (0.0008)
contract2 (1-yr)	−0.1589*** (0.0131)	−0.5713*** (0.0571)
contract3 (2-yr)	−0.1872*** (0.0133)	−1.1753*** (0.0883)
paperless	0.0717*** (0.0098)	0.2785*** (0.0418)
Observations	7,043	7,043

Part (b): Interpreting LPM Coefficients

1. Continuous Regressor: Monthly Charges ($b_5 = 0.0032$)

$$\frac{\widehat{\partial \text{churn}}}{\partial \text{monthlycharges}} = b_5 = 0.0032 \quad (+0.32 \text{ percentage points})$$

Interpretation: A **\$1 increase** in monthly charges is associated with an average increase of **0.32 percentage points** in the probability of switching telecom provider, ceteris paribus.

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2. Binary Regressor: Paperless Billing ($b_8 = 0.0717$)

$$\widehat{\Delta \text{churn}} = b_8 = 0.0717 \quad (+7.17 \text{ percentage points})$$

Interpretation: On average, all else equal, a customer who receives an electronic bill is **7.17 percentage points more likely** to switch providers than one receiving a paper bill.

Part (c): Tenure Turning Point Derivation

How does tenure affect churn?

$$\widehat{\text{churn}}_i = \dots + b_3 \text{tenure}_i + b_4 \text{tenure}_i^2 + \dots$$

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$$\frac{\partial \Pr(\text{churn}_i = 1)}{\partial \text{tenure}_i} = b_3 + 2b_4 \text{tenure}_i$$

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$$b_3 + 2b_4 \text{tenure}^* = 0 \implies \text{tenure}^* = -\frac{b_3}{2b_4}$$

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Step 3: Plug in LPM coefficient estimates:

$$\text{tenure}^* = -\frac{-0.0119}{2 \times 0.0001} \approx \mathbf{59.5} \text{ months}$$

Part (c): Tenure Turning Point — Delta Method & Economic M

Precision via the Delta Method in R

```
deltaMethod(t7_eq1, "-b3/(2*b4)", vcov=cov_t7_eq1)
```

Estimate: 59.95 months **SE:** 2.58 **95% CI:** [54.90, 65.00]

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Geometric Shape & Interpretation:

- $b_3 < 0$ (starts negative) and $b_4 > 0$ (slope increases) $\implies$ **U-shaped curve.**
- Churn probability reaches its **minimum at ≈ 60 months (≈ 5 years).**

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Empirical Reality Check

Only **21%** of customers (1,483 / 7,043) have tenure > 60 months! For the vast majority, longer tenure strictly reduces churn.

Part (d): Diagnostic Check — LPM Predicted Probabilities

Summary of LPM Fitted Values $\hat{Y}_i = \mathbf{x}_i' \mathbf{b}$

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-0.1969	0.0983	0.2664	0.2654	0.4444	0.7137

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- **Probability Violation:** Minimum predicted probability is **-19.7%**! Probabilities must strictly lie in $[0, 1]$.
- **Constant Marginal Effects:** The effect of an extra \$10 bill is assumed identical for a customer at 5% risk vs. 90% risk.
- **Inherent Heteroskedasticity:** $\text{Var}(\varepsilon_i | \mathbf{x}_i) = P_i(1 - P_i)$, requiring robust SEs.

Part (e): The Probit Model Solution

Standard Normal CDF Link Function

$$\Pr(\text{churn}_i = 1 \mid \mathbf{x}_i) = \Phi(l_i) = \int_{-\infty}^{l_i} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz$$

where l_i is the linear **Index Function**:

$$l_i = \beta_0 + \beta_1 \text{partner}_i + \cdots + \beta_8 \text{paperless}_i$$

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► **Key Takeaway:** Both models produce nearly identical sample means ($\approx 26.6\%$), but Probit guarantees all probabilities are strictly bounded inside $(0, 1)$.

Concept Check: Why Can't We Read Probit Coefs Directly?

Voting Question

In Table 1, the probit coefficient on `monthlycharges` is $\beta_5 = 0.0152$.

Does this mean a \$1 increase in charges increases churn probability by 1.52 percentage points?

Concept Check: Why Can't We Read Probit Coefs Directly?

Voting Question

In Table 1, the probit coefficient on `monthlycharges` is $\beta_5 = 0.0152$. Does this mean a \$1 increase in charges increases churn probability by 1.52 percentage points?

Answer: NO!

- $0.0152 = \frac{\partial l_i}{\partial \text{monthlycharges}}$ is the effect on the **unobservable index** l_i , not probability!
- To get the effect on **probability**, we must apply the chain rule:

$$\frac{\partial \Pr(\text{churn}_i = 1)}{\partial x_k} = \underbrace{\phi(l_i)}_{\text{Normal PDF}} \times \beta_k$$

- The marginal effect depends on the customer's specific location on the S-curve!

Part (g): Probit Marginal Effect for Continuous Variable (Tenure)

Calculus Chain Rule with Quadratic Tenure

$$\frac{\partial \Pr(\text{churn}_i = 1)}{\partial \text{tenure}_i} = \phi(l_i) \times \frac{\partial l_i}{\partial \text{tenure}_i} = \phi(\mathbf{l}_i) \times (\beta_3 + 2\beta_4 \text{tenure}_i)$$

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How do we compute the Average Marginal Effect (AME) in R?

- 1 Predict index for each person: $\hat{l}_i = \mathbf{x}_i' \mathbf{b}$ (`predict(t7_eq2)`).
- 2 Compute standard normal PDF:
 $\phi(\hat{l}_i) = \text{dnorm}(\text{index_probit}, 0, 1)$.
- 3 Compute index slope: $b_3 + 2b_4 \times \text{tenure}_i$.
- 4 Multiply individual marginal effects: $\text{ME}_i = \phi(\hat{l}_i)(b_3 + 2b_4 \text{tenure}_i)$.
- 5 Take the sample average: $\text{AME} = \frac{1}{n} \sum_{i=1}^n \text{ME}_i = -\mathbf{0.0064}$.

Part (h): AME Interpretation & The 'margins' Package

Verification in R via margins

```
summary(margins(t7_eq2, variables = "tenure"))
```

AME(tenure) = -0.0064 (SE: 0.0004, $p < 0.001$)

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Economic Interpretation

All else equal, staying with one's current provider an extra month is estimated to reduce the probability of switching by **0.64 percentage points on average**.

Part (j): Marginal Effect for Dummy Regressors (Paperless)

Crucial Rule for Dummy Variables

You **cannot take derivatives** of binary dummies!

Must evaluate the **finite discrete difference**:

$$ME = \Pr(\text{churn} = 1 \mid \text{paperless} = 1, \mathbf{x}) - \Pr(\text{churn} = 1 \mid \text{paperless} = 0, \mathbf{x})$$

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Profile: partner=1, dependents=0, tenure=30, monthlycharges=\$70, contract2=0, contract3=0.

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$$l_1 = b_0 + b_1 + 30b_3 + 30^2b_4 + 70b_5 + b_8 = -\mathbf{0.4312} \implies \Phi(l_1) = \mathbf{0.3332}$$

$$l_0 = b_0 + b_1 + 30b_3 + 30^2b_4 + 70b_5 = -\mathbf{0.7096} \implies \Phi(l_0) = \mathbf{0.2390}$$

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$$\text{ME}(\text{paperless}) = 0.3332 - 0.2390 = \mathbf{0.0942} \quad (+9.42 \text{ percentage points})$$

Part (k): Longhand vs. R 'margins' Nuance

Comparison for Paperless Billing at Specific Profile

- **Exact discrete difference (Longhand): 0.0942** (9.42 percentage points)
- **R margins output: 0.0952** (9.52 percentage points)

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Why is there a slight difference?

margins by default approximates marginal effects using numerical derivatives unless instructed. For binary variables, the exact finite difference $\Phi(l_1) - \Phi(l_0)$ is the gold standard!

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Interpretation: All else equal, this representative customer is **9.4 percentage points more likely to switch** if they receive electronic billing instead of paper statements.

Exam Practice Question (MCQ)

Question (Page 4)

The probit model is better than the linear probability model primarily because:

- ① it limits the estimated marginal effects to the $[0, 1]$ range
- ② it limits the predicted probabilities to the $[0, 1]$ range
- ③ both of the above
- ④ none of the above

Exam Practice Question (MCQ)

Question (Page 4)

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Suggested Answer: Option 2

Explanation: Probit transforms the linear index through the standard normal CDF $\Phi(I_i)$, strictly bounding predicted probabilities in $(0, 1)$. Marginal effects $\phi(I_i)\beta_k$ are not bounded in $[0, 1]$ (e.g., they can be negative or larger than 1 for continuous variables).

Key Takeaways: LPM vs. Probit

Feature	Linear Probability Model	Probit Model
Link Function	$P(Y = 1) = \mathbf{x}'\beta$	$P(Y = 1) = \Phi(\mathbf{x}'\beta)$
Bounds	Can fall outside $[0, 1]$	Strictly bounded in $(0, 1)$
Coefficient β_k	Direct marginal effect	Effect on index I_i
Marginal Effect	Constant (β_k) across individuals	Individual-specific: $\phi(I_i)\beta_k$
Dummy Regressor	Single slope estimate	Discrete change $\Phi(I_1) - \Phi(I_0)$
Estimation	OLS (Robust SEs)	Maximum Likelihood (MLE)